

8 Classical rate theory

For a large and complex reaction we cannot compute all the state-to-state reaction probabilities required to obtain the rate constant. However, one might guess that there is a more direct way to obtain the thermally-averaged rate constant. We shall discuss this approach within the context of classical mechanics but a similar argument also hold for the quantum case.

8.1 Linear response theory

In order to obtain a definition of the reaction rate, we perturb the system away from equilibrium and measure how it relaxes back to equilibrium when the perturbation is removed.

If our perturbed Hamiltonian is

$$H_\lambda = H - \lambda A \quad (8.1)$$

the initial thermal distribution will now be

$$\rho_\lambda = \frac{1}{Q_\lambda} e^{-\beta H_\lambda} \quad (8.2)$$

where the perturbed partition function Q_λ normalizes ρ_λ .

If at $t = 0$ we turn off the perturbation we can observe the resulting time-evolution of property B using

$$\langle B(t) \rangle_\lambda \equiv \int B(\vec{p}_t, \vec{q}_t) \rho_\lambda(\vec{p}, \vec{q}) d\vec{p} d\vec{q} \quad (8.3)$$

Unlike $\langle B(t) \rangle = \langle B \rangle$ which is time-independent (see lecture 2), $\langle B(t) \rangle_\lambda$ does depend on time as the dynamics are performed with the Hamiltonian, H , which is not the same Hamiltonian as the perturbed one used to define ρ_λ . The behaviour of $\langle B(t) \rangle_\lambda$ is rather complex in general but easier to study in the limit that $\lambda \rightarrow 0$, i.e. with a weak perturbation. Expanding to first order in λ gives

$$\langle B(t) \rangle_\lambda = \langle B \rangle + \beta \lambda [\langle A(0)B(t) \rangle - \langle A \rangle \langle B \rangle] + O(\lambda^2) \quad (8.4)$$

or using $\Delta A(t) = A(t) - \langle A \rangle$ and $\Delta B(t) = B(t) - \langle B \rangle$,

$$\langle \Delta B(t) \rangle_\lambda = \beta \lambda \langle \Delta A(0) \Delta B(t) \rangle + O(\lambda^2) \quad (8.5)$$

This powerful equation, known as the ‘Onsager regression hypothesis’ or ‘fluctuation-dissipation theorem’ allows us to express macroscopic relaxation of non-equilibrium systems in terms of spontaneous microscopic fluctuations in an equilibrium system.

8.2 Connection with macroscopic rate theory

Consider the following chemical reaction described by first-order rate constants



where k_+ is the forward rate constant and k_- is that for the reverse reaction.

Let $c_A(t)$ and $c_B(t)$ denote the concentrations then the phenomenological rate equations are

$$\frac{\partial c_A}{\partial t} = -k_+ c_A(t) + k_- c_B(t) \quad (8.7a)$$

$$\frac{\partial c_B}{\partial t} = k_+ c_A(t) - k_- c_B(t). \quad (8.7b)$$

Note that the total concentration of species, $c = c_A(t) + c_B(t)$, is constant.

At equilibrium, the concentrations $\langle c_A \rangle$ and $\langle c_B \rangle$ are stationary and thus $0 = -k_+ \langle c_A \rangle + k_- \langle c_B \rangle$ and the equilibrium constant is

$$K_{\text{eq}} \equiv \frac{\langle c_B \rangle}{\langle c_A \rangle} = \frac{k_+}{k_-}. \quad (8.8)$$

This relation between the forward and backward rate constant is known as “detailed balance”.

Defining $\Delta c_A(t) = c_A(t) - \langle c_A \rangle$ and $\Delta c_B(t) = c_B(t) - \langle c_B \rangle$, and hence $\Delta c_B(t) = -\Delta c_A(t)$, we find

$$\frac{\partial \Delta c_A}{\partial t} = -(k_+ + k_-) \Delta c_A(t) \quad (8.9a)$$

$$\frac{\partial \Delta c_B}{\partial t} = -(k_+ + k_-) \Delta c_B(t). \quad (8.9b)$$

These equations are solved by

$$\Delta c_A(t) = \Delta c_A(0) e^{-(k_+ + k_-)t} \quad (8.10a)$$

$$\Delta c_B(t) = \Delta c_B(0) e^{-(k_+ + k_-)t}. \quad (8.10b)$$

We therefore define the timescale for the reaction as $\tau = (k_+ + k_-)^{-1}$.

In this way we can relate the rate constants k_+ and k_- to the macroscopic relaxation of the concentration of reactants or products.

Multiplying and dividing equation (8.9b) by $\langle c_B \rangle$ and using the detailed-balance condition gives

$$\frac{\partial \Delta c_B}{\partial t} = -\frac{k_+ \langle c_B \rangle + k_- \langle c_A \rangle}{\langle c_B \rangle} \Delta c_B(t) \quad (8.11)$$

$$= -k_+ \frac{c}{\langle c_B \rangle} \Delta c_B(t). \quad (8.12)$$

We define the properties $A(\vec{p}, \vec{q})$ and $B(\vec{p}, \vec{q})$ from the previous section in such a way so as to determine whether the system is in a reactant or product state. They are projection operators returning a value between 0 and 1 and $A + B = 1$, although their exact form is not important as long as they successfully identify the reactant and product basins of attraction. Again we use A to permute the equilibrium and then allow the system to relax. This permutation will create more reactants and fewer products than would be present at equilibrium and we expect that the reactant concentration will decay back to its equilibrium value.

We can therefore identify $\langle A(t) \rangle_\lambda = c_A(t)/c$ and $\langle B(t) \rangle_\lambda = c_B(t)/c$. Using $\Delta B(t) = B(t) - \langle B \rangle$, $\dot{B}(t) \equiv \frac{\partial B}{\partial t} = \frac{\partial \Delta B}{\partial t} \equiv \Delta \dot{B}(t)$ and $\Delta A(t) = -\Delta B(t)$, we find

$$\langle \Delta \dot{B}(t) \rangle_\lambda = -k_+ \langle \Delta B(t) \rangle_\lambda / \langle B \rangle. \quad (8.13)$$

Rearranging gives

$$k_+ = -\frac{\langle \Delta \dot{B}(t) \rangle_\lambda}{\langle \Delta B(t) \rangle_\lambda} \langle B \rangle \quad (8.14)$$

$$= -\frac{\langle \Delta A(0) \Delta \dot{B}(t) \rangle}{\langle \Delta A(0) \Delta B(t) \rangle} \langle B \rangle \quad (8.15)$$

$$= \frac{\langle \Delta B(0) \Delta \dot{B}(t) \rangle}{\langle \Delta A(0) \Delta B(t) \rangle} \langle B \rangle \quad (8.16)$$

For a slow reaction, very few reactants starting in A will arrive in B and therefore $\langle \Delta A(0) \Delta B(t) \rangle = \langle A(0) B(t) \rangle - \langle A \rangle \langle B \rangle \approx -\langle A \rangle \langle B \rangle$ such that

$$k_+ = -\frac{\langle B(0) \dot{B}(t) \rangle}{\langle A \rangle} \quad (8.17)$$

$$= \frac{\langle \dot{B}(0) B(t) \rangle}{\langle A \rangle} \quad (8.18)$$

In reality the dynamical equations governing the molecular motion is more complicated than suggested by equation (8.7) and at short times there may be considerable deviation. The rate constant defined by equation (8.18) would then be time-dependent. However, at longer times it seems that many reactions do obey these rate laws (otherwise we would not have developed the concept of a rate constant at all). Therefore, we implicitly only apply this equation for values of t within a plateau region where k_+ is constant.

As we shall show in the next section, this equation is the basis of the flux correlation function formalism for computing rate constants.

8.3 Flux correlation functions

Let us introduce the reaction coordinate $s(\vec{q})$ and the momentum in this direction, $p_s(\vec{q})$, which describes the progress of the reaction. For example it could report on the bond distance during a chemical reaction in which the bond is broken. Our projection operators will be defined as $A = \Theta(s^\ddagger - s)$ and $B = \Theta(s - s^\ddagger)$ where Θ is the Heaviside step function and $s^\ddagger$ is a constant. When $s < s^\ddagger$, the configuration is a “reactant” and when $s > s^\ddagger$, the configuration describes a “product.”

The time-derivative of B is easily obtained using the chain rule

$$\dot{B} \equiv \frac{\partial B}{\partial t} = \frac{\partial B}{\partial \vec{q}} \cdot \frac{\partial \vec{q}}{\partial t} = \delta(s - s^\ddagger) \frac{\partial s}{\partial \vec{q}} \cdot \frac{\vec{p}}{m} \quad (8.19)$$

This is the flux through the dividing surface $s = s^\ddagger$ and can also be written as

$$\dot{B} \equiv F(p_s, s) = \delta[s - s^\ddagger] \frac{p_s}{m_s} \quad (8.20)$$

where p_s is the momentum in the s -direction with associated mass m_s .

The classical rate constant is therefore given by

$$k_{\text{cl}}(T) = Q_{\text{r}}(T)^{-1} C_{\text{fs}}^{\text{cl}}(t) \quad \text{for } t \text{ in plateau} \quad (8.21)$$

where $Q_r(T)$ is the reactant partition function

$$Q_r(T) = (2\pi\hbar)^{-3N} \iint e^{-\beta H(\vec{p}, \vec{q})} \Theta(s^\ddagger - s) d\vec{p} d\vec{q} \quad (8.22)$$

and the classical flux-side correlation function is

$$C_{fs}^{cl}(t) = (2\pi\hbar)^{-3N} \iint e^{-\beta H(\vec{p}, \vec{q})} F(p_s, s) \Theta[s(\vec{q}_t) - s^\ddagger] d\vec{p} d\vec{q} \quad (8.23)$$

where $\vec{q}_t \equiv \vec{q}_t(\vec{p}, \vec{q})$ is found by solving for the classical trajectory at time t with initial conditions $(\vec{p}, \vec{q})$.

The flux-side correlation function is the time-derivative of the side-side correlation function

$$C_{ss}^{cl}(t) = (2\pi\hbar)^{-3N} \iint e^{-\beta H(\vec{p}, \vec{q})} \Theta[s^\ddagger - s(\vec{q})] \Theta[s(\vec{q}_t) - s^\ddagger] d\vec{p} d\vec{q} \quad (8.24)$$

as can be shown using Liouville's theorem and the conservation of the Boltzmann operator. Therefore

$$k_{cl}(T) = Q_r(T)^{-1} \frac{d}{dt} C_{ss}^{cl}(t) \quad \text{for } t \text{ in plateau} \quad (8.25)$$

A very important point to note is that the rate constant is independent of the choice of the dividing surface. As long as the dividing surface separates reactants from products, it can have any shape or form. The dividing surface should always be placed in a transition region between reactants and products where the thermal distribution is negligible such that $Q_r(T)$ is independent. We can then prove

$$\frac{\partial}{\partial s^\ddagger} C_{ss}^{cl}(t) = (2\pi\hbar)^{-3N} \iint e^{-\beta H(\vec{p}, \vec{q})} \{ \delta[s^\ddagger - s(\vec{q})] \Theta[s(\vec{q}_t) - s^\ddagger] - \Theta[s^\ddagger - s(\vec{q})] \delta[s(\vec{q}_t) - s^\ddagger] \} d\vec{p} d\vec{q} \quad (8.26)$$

$$= (2\pi\hbar)^{-3N} \left\{ \iint e^{-\beta H(\vec{p}_0, \vec{q}_0)} \delta[s^\ddagger - s(\vec{q}_0)] \Theta[s(\vec{q}_t) - s^\ddagger] d\vec{p}_0 d\vec{q}_0 \right. \\ \left. - \iint e^{-\beta H(\vec{p}_0, \vec{q}_0)} \Theta[s^\ddagger - s(\vec{q}_0)] \delta[s(\vec{q}_t) - s^\ddagger] d\vec{p}_0 d\vec{q}_0 \right\} \quad (8.27)$$

$$= (2\pi\hbar)^{-3N} \left\{ \iint e^{-\beta H(\vec{p}_0, \vec{q}_0)} \delta[s^\ddagger - s(\vec{q}_0)] \Theta[s(\vec{q}_t) - s^\ddagger] d\vec{p}_0 d\vec{q}_0 \right. \\ \left. - \iint e^{-\beta H(\vec{p}_t, \vec{q}_t)} \Theta[s^\ddagger - s(\vec{q}_0)] \delta[s(\vec{q}_t) - s^\ddagger] d\vec{p}_t d\vec{q}_t \right\} \quad (8.28)$$

$$= (2\pi\hbar)^{-3N} \left\{ \iint e^{-\beta H(\vec{p}_0, \vec{q}_0)} \delta[s^\ddagger - s(\vec{q}_0)] \Theta[s(\vec{q}_t) - s^\ddagger] d\vec{p}_0 d\vec{q}_0 \right. \\ \left. - \iint e^{-\beta H(\vec{p}_0, \vec{q}_0)} \Theta[s^\ddagger - s(\vec{q}_{-t})] \delta[s(\vec{q}_0) - s^\ddagger] d\vec{p}_0 d\vec{q}_0 \right\} \quad (8.29)$$

$$= (2\pi\hbar)^{-3N} \iint e^{-\beta H(\vec{p}_0, \vec{q}_0)} \delta[s^\ddagger - s(\vec{q}_0)] \{ 2\Theta[s(\vec{q}_t) - s^\ddagger] - 1 \} d\vec{p}_0 d\vec{q}_0 \quad (8.30)$$

where we have explicitly shown the time of the initial conditions using $\vec{q}_0 \equiv \vec{q}$ and $\vec{p}_0 \equiv \vec{p}$. Therefore

$$\frac{\partial}{\partial s^\ddagger} C_{\text{fs}}^{\text{cl}}(t) = \frac{\partial}{\partial t} \frac{\partial}{\partial s^\ddagger} C_{\text{ss}}^{\text{cl}}(t) \quad (8.31)$$

$$= 2(2\pi\hbar)^{-3N} \iint e^{-\beta H(\vec{p}_0, \vec{q}_0)} \delta[s^\ddagger - s(\vec{q}_0)] \frac{p_s(\vec{p}_t, \vec{q}_t)}{m_s} \delta[s(\vec{q}_t) - s^\ddagger] d\vec{p}_0 d\vec{q}_0 \quad (8.32)$$

$$= 2(2\pi\hbar)^{-3N} \iint e^{-\beta H(\vec{p}_t, \vec{q}_t)} \delta[s^\ddagger - s(\vec{q}_0)] \frac{p_s(\vec{p}_t, \vec{q}_t)}{m_s} \delta[s(\vec{q}_t) - s^\ddagger] d\vec{p}_t d\vec{q}_t \quad (8.33)$$

$$= 2(2\pi\hbar)^{-3N} \iint e^{-\beta H(\vec{p}_0, \vec{q}_0)} \delta[s^\ddagger - s(\vec{q}_{-t})] \frac{p_s(\vec{p}_0, \vec{q}_0)}{m_s} \delta[s(\vec{q}_0) - s^\ddagger] d\vec{p}_0 d\vec{q}_0 \quad (8.34)$$

$$= -2(2\pi\hbar)^{-3N} \iint e^{-\beta H(\vec{p}, \vec{q})} \delta[s^\ddagger - s(\vec{q}_t)] \frac{p_s(\vec{p}, \vec{q})}{m_s} \delta[s(\vec{q}) - s^\ddagger] d\vec{p} d\vec{q} \quad (8.35)$$

The derivative of $C_{\text{fs}}^{\text{cl}}(t)$ with respect to $s^\ddagger$ obviously tends to 0 for large times as it would be extremely unlikely for a trajectory to return to the dividing surface after time t . The rate is therefore independent of the dividing surface.

8.4 Classical transition-state theory

Transition-state theory is one of the most important concepts in chemistry and can be rigorously understood in terms of the classical flux-side correlation function.

We know that the exact classical rate is obtained from the plateau region of $C_{\text{fs}}^{\text{cl}}(t)$ but to compute that would involve propagating chaotic trajectories on the PES which is difficult to study analytically. Instead TST is defined from the short-time limit and is much easier to study although one must bear in mind that the short-time limit may or may not be in the plateau region which greatly affects the accuracy of the approximation.

$$k_{\text{TST}}(T) = Q_{\text{r}}(T)^{-1} \lim_{t \rightarrow 0_+} C_{\text{fs}}^{\text{cl}}(t) \quad (8.36)$$

This limit can be obtained by linearizing the dynamics, $\vec{q}_t \approx \vec{q}_0 + \frac{\vec{p}_0}{m} t$, which is exact in the short-time limit:

$$\lim_{t \rightarrow 0_+} F(p_s, s) \Theta[s(\vec{q}_t) - s^\ddagger] = \lim_{t \rightarrow 0_+} F(p_s, s) \Theta \left[s^\ddagger + \frac{p_s}{m_s} t - s^\ddagger \right] \quad (8.37)$$

$$= \lim_{t \rightarrow 0_+} F(p_s, s) \Theta[p_s] \quad (8.38)$$

Note that the classical flux-side correlation function is discontinuous at $t = 0$ and gives the negative result for $t \rightarrow 0_-$.

This gives the classical TST rate

$$k_{\text{TST}}(T) = (2\pi\hbar)^{-3N} Q_{\text{r}}(T)^{-1} \iint e^{-\beta H(\vec{p}, \vec{q})} \delta(s - s^\ddagger) \frac{p_s}{m_s} \Theta(p_s) d\vec{p} d\vec{q} \quad (8.39)$$

$$= (2\pi\hbar)^{-3N} Q_{\text{r}}(T)^{-1} \int e^{-\beta \vec{p}^2 / 2m} \frac{p_s}{m_s} \Theta(p_s) d\vec{p} \int e^{-\beta V(\vec{q})} \delta(s - s^\ddagger) d\vec{q}. \quad (8.40)$$

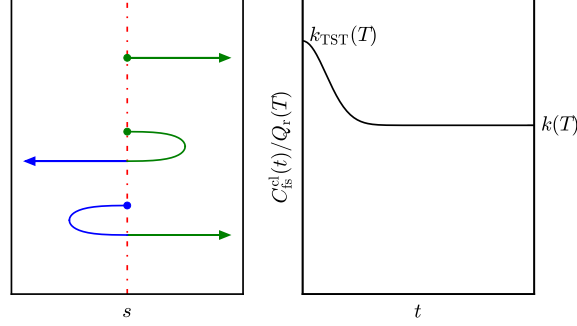


Figure 7: Trajectories starting at the dividing surface contribute (green) to the TST rate at short times and to the full rate at long times. The two lower trajectories recross and reduce the full rate.

Using $Q_r = (2\pi\hbar)^{-3N} \iint e^{-\beta H(\vec{p}, \vec{q})} \Theta(s^\ddagger - s) d\vec{p} d\vec{q}$, and $\langle O \rangle = \frac{\iint e^{-\beta H(\vec{p}, \vec{q})} O(\vec{p}, \vec{q}) d\vec{p} d\vec{q}}{\iint e^{-\beta H(\vec{p}, \vec{q})} d\vec{p} d\vec{q}}$, we find

$$k_{\text{TST}}(T) = \frac{1}{2} \langle |\dot{s}| \rangle \frac{\langle \delta(s - s^\ddagger) \rangle}{\langle \Theta(s^\ddagger - s) \rangle} \quad (8.41)$$

$$= \frac{1}{2} \langle |\dot{s}| \rangle e^{-\beta \Delta A} \quad (8.42)$$

where

$$\langle |\dot{s}| \rangle = \frac{\int e^{-\beta \vec{p}^2/2m} |p_s/m_s| d\vec{p}}{\int e^{-\beta \vec{p}^2/2m} d\vec{p}} \quad (8.43)$$

and

$$\Delta A = -\frac{1}{\beta} [\ln \langle \delta(s - s^\ddagger) \rangle - \ln \langle \Theta(s^\ddagger - s) \rangle] \quad (8.44)$$

is the free-energy difference between the dividing surface and the reactants.

This gives us the simple Arrhenius picture for chemical reaction rates, in that they depend exponentially on free-energy barriers. The prefactor is an easily-obtained flux term which depends only on the dividing-surface shape and requires no evaluations of the PES.

However, TST does not always give the same result as the full classical rate as recrossing is neglected. A consideration of the possible recrossing trajectories (Figure 7) shows that recrossing will always lower the rate such that we can say that TST is an upper bound to the full classical rate.

Note that although the full rate is not dependent on the dividing surface, the TST rate is strongly dependent. The variational principle allows us to define an optimal TST with the dividing surface chosen so as to minimize the TST rate. This optimal TST is the best estimate for the full rate.

It is hard to estimate the effect that recrossing will have on the rate without computing many trajectories and observing their long-time behaviour. However, if we did know it, we could write,

$$k_{\text{cl}}(T) = \nu k_{\text{TST}}(T) \quad (8.45)$$

where $0 < \nu < 1$ is the recrossing factor.

8.5 Quantum flux correlation functions

Much of the theory derived above can be rigorously extended to describe quantum rate theory in terms of the quantum correlation functions

$$k(T) = Q_r(T)^{-1} \frac{d}{dt} C_{ss}(t) = Q_r(T)^{-1} C_{fs}(t) = Q_r(T)^{-1} \int_0^t C_{ff}(t') dt' \quad (\text{for } t \text{ in plateau}) \quad (8.46)$$

where

$$Q_r = \text{Tr} \left[e^{-\beta \hat{H}} \Theta(\hat{s} - s^\dagger) \right] \quad (8.47)$$

$$C_{ss}(t) = \text{Tr} \left[e^{-\beta \hat{H}/2} \Theta(s^\dagger - \hat{s}) e^{-\beta \hat{H}/2} e^{i\hat{H}t/\hbar} \Theta(\hat{s} - s^\dagger) e^{-i\hat{H}t/\hbar} \right] \quad (8.48)$$

$$C_{fs}(t) = \text{Tr} \left[e^{-\beta \hat{H}/2} \hat{F} e^{-\beta \hat{H}/2} e^{i\hat{H}t/\hbar} \Theta(\hat{s} - s^\dagger) e^{-i\hat{H}t/\hbar} \right] \quad (8.49)$$

$$C_{ff}(t) = \text{Tr} \left[e^{-\beta \hat{H}/2} \hat{F} e^{-\beta \hat{H}/2} e^{i\hat{H}t/\hbar} \hat{F} e^{-i\hat{H}t/\hbar} \right] \quad (8.50)$$

where the flux operator, $\hat{F}$, was defined in the previous chapter, equation (7.6). Here we symmetrized the thermal density around one of the operators, which makes the functions real, but isn't actually necessary in order to obtain the correct rate constant.

For a free particle, $H = \hat{p}^2/2m$, it is possible to find an analytic expression for the quantum flux-flux correlation function, equation (8.50). By expanding the trace in momentum states, show that

$$C_{ff}(t) = \frac{1}{2\pi\beta\hbar} \frac{(\hbar\beta/2)^2}{[t^2 + (\hbar\beta/2)^2]^{3/2}}.$$

Hence plot $C_{ss}(t)$, $C_{fs}(t)$ and $C_{ff}(t)$. Sketch the equivalent functions from classical mechanics. Compute the quantum-mechanical rate constant and show that classical rate theory gives the same result for this particular case.

Unlike the classical flux-side correlation function, $C_{fs}(t)$ is continuous at $t = 0$ and thus one cannot simply define a quantum TST by taking the short-time limit. Finding a quantum TST is an ongoing active area of research in our group and elsewhere.

8.6 Further Reading

The following textbooks will give you further information on the topics of this lecture:

Linear response theory and classical rate

- Chandler, Introduction to Modern Statistical Mechanics (Oxford), chapter 8

Linear response theory and diffusion constants

- Tuckerman, Statistical Mechanics (Oxford), chapter 13

Hopefully you may be able to read them on the internet.